ADR. In this case, it is not an ADR, because taking Prozac was the treatment for reducing anxiety. Let us consider another example.

E2: "After starting to use Valium, I started getting tension headaches. I was told that I may get stomach pains and acidity, but that never happened during all the months I was on Valium."

In this example, the ADR present is <*Valium, tension headache*>. While stomach pains and acidity are potential side effects, the sentence contains an implicit negation ('that never happened') concerning these side effects. Hence, they should not be considered as ADRs in this case.

Let us consider one more example.

E3: "The patient was administered Heparin to reduce the risks of clotting, but this led to an allergic reaction and intestinal bleeding."

This example contains two ADRs namely, <*Heparin, allergic reaction*> and <*Heparin, intestinal bleeding*>. Also note that unlike the two earlier examples which were patient reported ADRs, this is an example of the ADR present in a clinical text. Hence the language is more complex and contains medical terminology. The point is that the vocabulary of texts containing ADRs can vary considerably, depending on the source of the data.

The problem of detecting ADR from text can be approached in multiple ways. The simplest approach is to treat it as a text classification-problem first. Given a document, we will first classify each sentence based on whether it contains an ADR or not. This can be modelled as a binary classification problem. This is the task of ADR sentence classification. We can employ standard text classification methods such as Support Vector Machines, Random Forests, or even deep learning based text classifiers for solving this task.

Once we have the sentences that are marked as containing an ADR, we can then pass them through another system which extracts the actual ADRs from them. ADR extraction can be modelled in multiple ways. We can treat it as a sequence-tagging/labelling problem. The tags can be <*B_ADR, I_ADR* and *O_ADR*> where *B_ADR* is the tag representing the beginning of the ADR, *I_ADR* is the tag indicating that the word is inside an ADR, and *O_ADR* is the tag indicating that the word is outside any ADR. The task is to label each word in the sentence with an associated tag. Another way of modelling the ADR extraction problem is to treat it as a relation-extraction problem. The first step is to do entity-identification, wherein the entities of interest are the drugs' names and their respective side effects.

Once we have identified the entities, the next step is to do relation-extraction, wherein we want to identify that the relation <*Entity1* is associated with *Entity2*> (where Entity1 is a drug and Entity2 is a side effect). There are well known techniques available for relation extraction using both classical machine learning methods as well as deep learning methods. Any of these techniques can be employed.

It is also possible to approach the problem of ADR extraction directly using any of the above mentioned methods instead of first classifying each sentence as ADR relevant or not. However, given large documents where many of the sentences may not be relevant to ADR extraction, it is better to do ADR relevant sentence classification first before performing ADR extraction directly on the set of sentences classified as relevant, right from the first step.

So far, we have been assuming that ADR is typically present within a sentence. Consider the following example.

E4: "I have been feeling great. One thing though, regarding my sleep; it's been disturbed since I started on Prozac."

This is an example where the ADR is present across sentences. Hence this would need inter-sentence analysis to extract the ADR. Hence, a simple sequence-labelling technique at the sentence level will not suffice for this example. Cross-sentence relation extraction is needed for ADR extraction in this case.

What are the data sets available for ADR extraction? If you want to consider the ADR relevant sentence binary classification task, you can use the data set for Task 3 available at *https://healthlanguageprocessing.org/smm4h/social-media-mining-for-health-applications-smm4h-workshop-shared-task/*. Another data set is the CSIRO Adverse Drug Event Corpus (CADEC) available from *https://data.csiro.au/dap/landingpage?pid=csiro:10948&v=3&d=true*. A number of papers have used the CADEC corpus, which is a rich annotated corpus of medical forum posts on patient reported ADEs. This can be used for ADR extraction.

Of late, a number of deep learning techniques have been applied to this problem of detecting ADRs. An interesting approach to ADR extraction is discussed in the paper, 'An Attentive Sequence Model for Adverse Drug Event Extraction from Biomedical Text', available at *https://arxiv.org/abs/1801.00625*. In this paper, the problem is framed in the question-answer format. We will discuss some of these techniques in next month's column. Meanwhile, I would encourage our readers to download and experiment with the data sets mentioned above.

If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at *sandyasm_AT_yahoo_DOT_com*. Wishing all our readers happy coding until next month!

By: Sandya Mannarswamy

The author is an expert in systems software and is currently working as a research scientist at Conduent Labs India (formerly Xerox India Research Centre). Her interests include compilers, programming languages, file systems and natural language processing. If you are preparing for systems software interviews, you may find it useful to visit Sandya's LinkedIn group 'Computer Science Interview Training India' at *http://www.linkedin.com/groups?home=&gid=2339182*.